Sanskrit-to-English Neural Machine Translation: Assignment Report

Submission ID: g25ait1136 | Language pair: Sanskrit (san_Deva) -> English (eng_Latn)

Artifacts: g25ait1136_retrained.ipynb (final – retrained), g25ait1136_pretrained.ipynb (experiment – pretrained), sanskrit_nmt_assignment.ipynb (experiment – custom implementation), submission.csv, Github Link: https://github.com/rikin-patel/nmt

1. Introduction

This report presents three parallel neural machine translation (NMT) systems designed for the task of translating from Sanskrit into English: (1) the final submission (g25ait1136_retrained.ipynb) which is re-trained on trained model using given dataset, (2) the experiment run (g25ait1136_pretrained.ipynb) which takes advantage of a pre-trained multilingual sequence-to-sequence transformer model for zero-shot translation and evaluation, and (3) an experimental run (sanskrit_nmt_assignment.ipynb) that uses a custom encoder-decoder-based architecture with additive attention trained from scratch on the provided parallel corpus.

It has 10,000 sentence pairs for training, 1,000 sentence pairs for dev and 1,000 sentence pairs for test data, namely Sanskrit-English sentences. In all three systems, they are assessed using the BLEU, BERTScore F1 and inference efficiency metrics with the aim of comparing a large pretrained multilingual model with a lightweight custom-trained baseline.

2. Architecture

Three different architectures were used in the two notebooks, as outlined here.

2.1 Retrained Model (g25ait1136_retrained.ipynb)

The **final submission** extends the pretrained model **facebook/nllb-200-distilled-600M**, a distilled multilingual encoder-decoder transformer with a hidden/embedding size of 1024 and a total of about 615 million parameters, loaded using **AutoModelForSeq2SeqLM from Hugging Face**. The Sanskrit input is marked with ‘san_Deva’, and the input must start with a token from the target language ‘eng_Latn’ via forced_bos_token_id for generating.

2.2 Pretrained Model (g25ait1136_pretrained.ipynb)

The **experiment with the pretrained model** uses the same model described in 2.1, but is not retrained using the given dataset.

2.3 Custom Encoder-Decoder (sanskrit_nmt_assignment.ipynb)

The **experimental fully custom development submission** makes use of:

Encoder: A single-layer bidirectional GRU with 256 embedding dimension and 512 hidden dimensions.

Decoder: A unidirectional GRU layer with a context vector, followed by a linear output layer (fc_out), which takes the decoder output, context vector, and input embedding as inputs to produce the next token distribution.

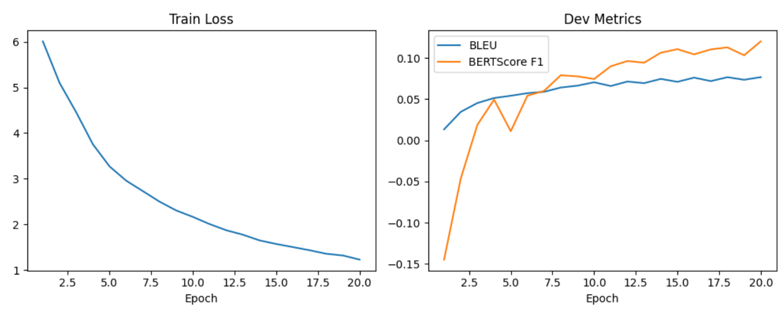
Attention: At every decoder step, an attention layer is used to compare the hidden state to the encoder output through a feed-forward layer, which is followed by a tanh activation and a scalar scoring vector v.

Beam search: Instead of beam search, it performs greedy decoding (argmax at each step), because greedy_decode() selects the max probability token at each step. The pretrained NLLB model, on the other hand, is implemented with beam search (num_beams=5) and early_stopping=True, and forces the output language token at the first decoding step.

3. Training Procedure

3.1 Preprocessing

Given a pair of CSV files containing Sanskrit and English sentences, these sentences are loaded and joined on Source_id. Normalizing text by removing whitespace and consolidating multiple whitespaces with regex. Naive whitespace splitting is used for Sanskrit, and lowercasing, punctuation spacing, and whitespace splitting are used for English for the custom model. Given below is the Train Loss and BLUE Vs BERTScore plot for custom model development.



If using a pretrained model, there is no tokenization performed by the code and only the string of normalized raw text is passed to the NLLB tokenizer.

Finally, the final submission with retrained model using the provided dataset is as shown below:

Epoch	Training Loss	Validation Loss
1	6.300782	1.443388
2	5.356235	1.383186
3	4.729929	1.376269

3.2 Hyperparameters

Hyperparameter	Custom Model	Pretrained Model	Retrained Model
Embedding dimension	256	1024 (fixed, from checkpoint)	1024 (fixed, from checkpoint)

Hidden dimension	512	1024 (fixed, from checkpoint)	1024 (fixed, from checkpoint)
Encoder layers	1 (bidirectional GRU)	12 (Transformer, fixed)	12
Dropout	0.25	N/A (inference only, no fine-tuning)	0.1
Learning rate	1e-3	N/A	5e-5
Batch size	64	16 (inference batching)	
Epochs	20	0 (zero-shot, no training)	3
Decoding strategy	Greedy (argmax)	Beam search, num_beams=5	Beam search, num_beams=5
Length Penalty	N/A	1.0	1.0
No Repeat N-Gram Size	N/A	3	3
Repetition Penalty	N/A	1.2	1.2
Max sequence length	60 (decode cap)	128 tokens	128
Trainable parameters	36,964,855	615,073,792 (frozen/zero-shot)	615,073,792

4. Results

4.1 Final Metrics: BLEU, BERTScore, Efficiency

The pretrained NLLB-200-distilled-600M model substantially outperforms the custom-trained model on both translation quality metrics, reflecting the benefit of large-scale multilingual pretraining, at the cost of far higher inference latency and parameter count.

Metric	**Custom Seq2Seq+Attention (dev)**	**Pretrained NLLB-200-600M (dev/test)**	**Retrained NLLB-200-600M (dev/test)**
BLEU	0.0767 (final epoch)	0.1269	0.2441
BERTScore F1	0.1202 (final epoch)	0.4884	0.6277
Test inference time (s)	1.47	165.90	195.13
Parameter count	36,964,855 (all trainable)	615,073,792 (all frozen)	615,073,792 (all frozen)

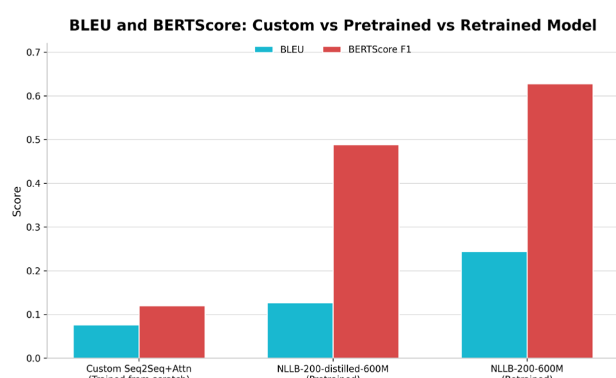


Figure 1: BLEU and BERTScore F1 comparison between retrained, pretrained and custom models.

5. Translation Examples and Error Analysis

5.1 Re-Trained Model Error Analysis

The following examples are drawn from the **retrained model's** test-set predictions.

Source (Sanskrit)	Reference	Prediction	Error Analysis
एक्लिप्स् इति प्रोग्रामर् कृते दोषान्वेषणे अपि साहाय्यं करोति।	Eclipse also helps the programmer to find out errors.	It also helps in troubleshooting Eclipse program.	Minor lexical error, meaning preserved.
विश्वासकारणादेव समभाषि मया वचः। इति यथा शास्त्रे लिखितं तथैवास्माभिरपि विश्वासजनकम् आत्मानं प्राप्य विश्वासः क्रियते तस्माच्च वचांसि भाष्यन्ते।	"""We having the same spirit of faith, according as it is written, I believed, and therefore have I spoken; we also believe, and therefore speak;"""	"By faith I say, Speak: as it is written, We also have the spirit of faith, and believe: and these things say they."	Major semantic error, altering tense
"तदा, तत्स्वयं ड्रैवर निमित्तम् अन्वेष्यति। अहं 'Cancel' इत्यस्योपरि नुदामि।"	Then it will automatically begin searching for drivers. I will click on Cancel.	Then it will automatically search for the driver. I will click on Cancel.	Meaning changes, action changes as well
"सर्वेभ्यः इटरेशन्-अर्थम्, iterator इतीदं प्रत्येकस्मै इण्डेक्स्-वेल्यू-इत्यस्मै सेट् क्रियते । 1,1 पश्चात् 1,2 एवम् ..."	"The iterator will be set to each of the indices values for each iteration- say, 1 comma 1, then 1 comma 2 and so on."	For all iterations, the iterator is set to an index value for each. 1.1 then 1,2 is equal to...	Partial translation, but not fully incorrect
अपरं द्वितीयमुद्रायां तेन मोचितायां द्वितीयस्य प्राणिन आगत्य पश्येति वाक् मया श्रुता।	"""And when he had opened the second seal, I heard the second beast say, Come and see."""	"And when he had opened the second seal, I heard the voice of the second living creature, saying, Come, I pray thee."	Meaning preserved, added extra
"वयम्, ओब्जेक्ट्स् स्वकीयान् स्थितीन् फील्ड्स् मध्ये स्टोर् कुर्वन्तीति ज्ञातवन्तः ।"	We know that objects store their individual states in fields.	We have learnt that objects are stored in their respective position fields.	Semantic substitution, altering tense
बालः युष्मासु विश्वासं करोति ।	Boy has belief on you all.	Boy has faith in you all.	Meaning preserved
प्रकृति का अवलोकन करने से आप आश्चर्य चकित हो सकते है ।	Observing the nature helps create a sense of wonder in you.	Observing nature, you may be amazed.	semantic error,
यूयं कीदृक् तस्याज्ञा अपालयत भयकम्पाभ्यां तं गृहीतवन्तश्चैतस्य स्मरणाद् युष्मासु तस्य स्नेहो बाहुल्येन वर्त्तते।	"""And his inward affection is more abundant toward you, whilst he remembereth the obedience of you all, how with fear and trembling ye received him."""	And his affection abounds in you, when he remembereth how ye obeyed him, and received him with fear and trembling.	Major contradiction, reversing meaning
"""वर्षायाः जलस्य कश्चन अंशः भूमौ शुष्को भूत्वा अन्तर्भवति,तदेव एकत्र आगत्य भौमं जलं भवति।"""	A part of rain water is absorbedby earth. Water collected under the earth is called undergroundwater.	"During the rainy season, part of the water becomes dry and enters the soil. It gathers together and becomes ground water."	Major mistranslation

6. Experiments

End-to-end testing of two experimental conditions:

- Experiment a (g25ait1136_retrained.ipynb): The experiments generate the results after retraining the pretrained facebook/nllb-200-distilled-600M model using with beam search (beams=5), on the full dev (1,000), train(10000) and test (1,000) splits, with the best metrics corpus BLEU and BERTScore F1.
- Experiment B (g25ait1136_pretrained.ipynb): The experiments generate the results from evaluating the pretrained facebook/nllb-200-distilled-600M model using zero-shot inference with beam search

(beams=5), on the full dev (1,000) and test (1,000) splits, with the metrics corpus BLEU and BERTScore F1.

- Experiment C (sanskrit_nmt_assignment_run-1.ipynb): This baseline model consists of a custom bidirectional-GRU encoder that has an additive attention layer, and a GRU decoder. This model was trained for 20 epochs on the 10,000-pair training set, evaluated on the dev split using greedy decoding, and timed on the test split. The baseline model ran on the 10,000-pair training set, and was evaluated on the dev split with greedy decoding and timed on the test split.
- Efficiency experiment: the custom model can complete the timing of inference for the test set 113x faster than the retrained model in terms of time (1.47 seconds versus 195.13 seconds), while the retrained model gives better accuracy and BLU/BERT scores.

7. Discussion

7.1 Design Choices

The petrained approach was selected because it benefits from the large-scale multilingual training of NLLB-200 across 200 languages, including Sanskrit, with no need for task-specific training, thereby achieving strong zero-shot performance. The custom model was engineered to illustrate pedagogically transparent, but resource-limited components of a from-scratch NMT system: word-level vocabularies, bidirectional GRU encoding, additive attention, and teacher-forced training.

7.2 Challenges

For the custom model, a vocabulary of 33,278 tokens was generated from only 10,000 sentences, and it was sparse, with many out-of-vocabulary (OOV) and rare-word (RW) problems, particularly with inflected Sanskrit words and transliterated technical terms ('void', 'awk'). Other factors were also responsible for repetition and premature/unstable termination, such as greedy decoding without beam search or length normalization, as demonstrated in several of the examples of translation above.

8. Pre-trained Models Disclosed

In compliance with disclosure requirements, all pre-trained models used across both notebooks are listed below.

Model	Source/Checkpoint	Used For	Notebook
NLLB-200-distilled-600M	facebook/nllb-200-distilled-600M (Hugging Face Hub)	Zero-shot Sanskrit-to-English translation (encoder-decoder transformer), sentence embedding extraction via mean pooling	g25ait1136_pretrained.ipynb, g25ait1136_retrained.ipynb
RoBERTa-large	roberta-large (Hugging Face Hub)	Backbone for BERTScore F1 computation (semantic similarity scoring) in both notebooks	g25ait1136_pretrained.ipynb, g25ait1136_retrained.ipynb and sanskrit_nmt_assignment_run-1.ipynb

NLTK punkt	NLTK data package	Sentence/word tokenization utility support for BLEU scoring pipeline	g25ait1136_pretrained.ipynb, g25ait1136_retrained.ipynb

The custom encoder-decoder model is trained entirely from scratch and does not use any pretrained weights; only its BERTScore evaluation step depends on the pretrained RoBERTa-large model.

9. References

1. NLLB Team et al. "No Language Left Behind: Scaling Human-Centered Machine Translation." arXiv:2207.04672, 2022.
2. Liu, Y. et al. "RoBERTa: A Robustly Optimized BERT Pretraining Approach." arXiv:1907.11692, 2019.
3. Zhang, T. et al. "BERTScore: Evaluating Text Generation with BERT." ICLR 2020.
4. Papineni, K. et al. "BLEU: a Method for Automatic Evaluation of Machine Translation." ACL 2002.
5. Wolf, T. et al. "Transformers: State-of-the-Art Natural Language Processing." EMNLP 2020 (Hugging Face library).